

Rapido Booking Data Analysis Project

Part-1 : Import, Load CSV, Basic Analysis

Step-1 Install Library

```
pip install pandas numpy matplotlib
```

Step-2 Import Library

```
import pandas as pd  
import numpy as np  
import matplotlib.pyplot as plt  
import os
```

Step-3 Create Charts Folder

```
if not os.path.exists("charts"):  
    os.makedirs("charts")
```

Step-4 Load Dataset

```
df = pd.read_csv("rapido_booking.csv")
```

Step-5 Display First 10 Records

```
print("="*60)
print("FIRST 10 RECORDS")
print("="*60)

print(df.head(10))
```

Step-6 Display Last 10 Records

```
print("="*60)
print("LAST 10 RECORDS")
print("="*60)

print(df.tail(10))
```

Step-7 Dataset Information

```
print("="*60)
print("DATASET INFORMATION")
print("="*60)

print(df.info())
```

Step-8 Dataset Shape

```
print("="*60)
print("DATASET SHAPE")
print("="*60)
print(df.shape)
```

Output

(100,12)

Step-9 Dataset Columns

```
print("="*60)
print("COLUMN NAME")
print("="*60)

print(df.columns)
```

Step-10 Data Types

```
print("="*60)
print("DATA TYPES")
print("="*60)

print(df.dtypes)
```

Step-11 Missing Value

```
print("="*60)
print("MISSING VALUE")
print("="*60)
```

```
print(df.isnull().sum())
```

Step-12 Duplicate Records

```
print("="*60)
print("DUPLICATE RECORD")
print("="*60)
```

```
print(df.duplicated().sum())
```

Step-13 Remove Duplicate

```
df = df.drop_duplicates()
```

```
print("Duplicate Removed Successfully")
```

Step-14 Dataset Statistics

```
print("="*60)
```

```
print("STATISTICS")
```

```
print("="*60)
```

```
print(df.describe())
```

Step-15 Statistics Including Object Columns

```
print(df.describe(include='all'))
```

Step-16 NumPy Array

```
fare = np.array(df["Fare (₹)"])
```

```
print("="*60)
```

```
print("NUMPY ARRAY")
```

```
print("="*60)
```

```
print(fare)
```

Step-17 NumPy Operations

```
print("Total Fare :", np.sum(fare))  
print("Average Fare :", np.mean(fare))  
print("Maximum Fare :", np.max(fare))  
print("Minimum Fare :", np.min(fare))  
print("Median Fare :", np.median(fare))  
print("Standard Deviation :", np.std(fare))
```

Step-18 Pandas Operations

```
print("="*60)
```

```
print("PANDAS OPERATIONS")
```

```
print("="*60)
```

```
print("Total Revenue :", df["Fare (₹)"].sum())
```

```
print("Average Fare :", df["Fare (₹)"].mean())
```

```
print("Maximum Fare :", df["Fare (₹)"].max())
```

```
print("Minimum Fare :", df["Fare (₹)"].min())
```

```
print("Median Fare :", df["Fare (₹)"].median())
```

```
print("Mode Fare")
```

```
print(df["Fare (₹)"].mode())
```

```
print("Standard Deviation")
```

```
print(df["Fare (₹)"].std())
```

Step-19 Unique Values

```
print("Vehicle Types")
```

```
print(df["Vehicle Type"].unique())
```

```
print("Payment Modes")
```

```
print(df["Payment Mode"].unique())
```

```
print("Booking Status")
```

```
print(df["Booking Status"].unique())
```

Step-20 Number of Unique Values

```
print(df.nunique())
```

Output

Total Revenue

Average Fare

Maximum Fare

Minimum Fare

Median Fare

Mode Fare

Standard Deviation

Vehicle Types

Payment Modes

Booking Status

Part-1 Complete

Part-2 topics cover:

- Vehicle Wise Booking
- Booking Status Analysis
- Payment Mode Analysis
- Highest & Lowest Fare Booking
- Revenue by Vehicle
- Revenue by Captain
- Revenue by Payment
- Pickup & Drop Location Analysis
- Daily Revenue Analysis
- Completed, Cancelled, Ongoing Ride Analysis
- **Rapido Booking Data Analysis Project**

- **Part-2 :**
Data Analysis using Pandas
- **Assumption:** df use in Part-1.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

Step-21 Vehicle Wise Booking

```
print("="*60)
print("VEHICLE WISE BOOKING")
print("="*60)
```

```
vehicle_booking = df["Vehicle Type"].value_counts()

print(vehicle_booking)
```

Step-22 Vehicle Wise Percentage

```
• print("="*60)
  print("VEHICLE WISE PERCENTAGE")
  print("="*60)

  vehicle_percentage = round(
      (df["Vehicle Type"].value_counts()/len(df))*100,
      2
  )

  print(vehicle_percentage)
```

Step-23 Booking Status

```
print("="*60)
print("BOOKING STATUS")
print("="*60)

status = df["Booking Status"].value_counts()

print(status)
```

Step-24 Booking Status Percentage

```
status_percent = round(
    (status/len(df))*100,
    2
)

print(status_percent)
```

Step-25 Payment Mode

```
print("="*60)
print("PAYMENT  MODE")
print("="*60)
```

```
payment = df["Payment Mode"].value_counts()

print(payment)
```

Step-26 Payment Percentage

```
payment_percent = round(
    (payment/len(df))*100,
    2
)
```

```
print(payment_percent)
```

Step-27 Highest Fare Booking

```
print("="*60)
print("HIGHEST FARE BOOKING")
print("="*60)

highest = df[df["Fare (₹)"]==df["Fare (₹)"].max()]

print(highest)
```

Step-28 Lowest Fare Booking

```
print("="*60)
print("LOWEST FARE BOOKING")
print("="*60)

lowest = df[df["Fare (₹)"]==df["Fare (₹)"].min()]

print(lowest)
```

Step-29 Revenue By Vehicle

```
print("="*60)
print("REVENUE BY VEHICLE")
print("="*60)

vehicle_revenue = df.groupby(
    "Vehicle Type"
)["Fare (₹)"].sum()

print(vehicle_revenue)
```

Step-30 Average Fare By Vehicle

```
print("="*60)
print("AVERAGE FARE BY VEHICLE")
print("="*60)

avg_vehicle = df.groupby(
    "Vehicle Type"
)["Fare (₹)"].mean()

print(avg_vehicle)
```

Step-31 Maximum Fare By Vehicle

```
print("="*60)
print("MAXIMUM FARE BY VEHICLE")
print("="*60)

print(
df.groupby(
"Vehicle Type"
)["Fare (₹)"].max()
)
```

Step-32 Revenue By Captain

```
print("="*60)
print("REVENUE BY CAPTAIN")
print("="*60)
```

```
captain = df.groupby(
    "Captain Name"
) ["Fare (₹)"].sum()
```

```
print(captain)
```

Step-33 Average Fare By Captain

```
print("="*60)
print("AVERAGE FARE BY CAPTAIN")
print("="*60)

print(
df.groupby(
"Captain Name"
)["Fare (₹)"].mean()
)
```

Step-34 Total Trips By Captain

```
print("="*60)
print("TOTAL TRIPS BY CAPTAIN")
print("="*60)

print(
df["Captain Name"].value_counts()
)
```

Step-35 Revenue By Payment Mode

```
print("="*60)
print("REVENUE BY PAYMENT MODE")
print("="*60)
payment_revenue = df.groupby(
    "Payment Mode"
)["Fare (₹)"].sum()

print(payment_revenue)
```

Step-36 Average Fare By Payment

```
print("="*60)
print("AVERAGE FARE BY PAYMENT")
print("="*60)

print(
df.groupby(
    "Payment Mode"
) ["Fare (₹)"].mean()
)
```

Step-37 Pickup Location Analysis

```
print("="*60)
print("PICKUP LOCATION")
print("="*60)

pickup = df["Pickup Location"].value_counts()

print(pickup)
```

Step-38 Top 5 Pickup Location

```
print("="*60)
print("TOP 5 PICKUP")
print("="*60)

print(
pickup.head()
)
```

Step-39 Drop Location Analysis

```
print("="*60)
print("DROP LOCATION")
print("="*60)

drop = df["Drop Location"].value_counts()

print(drop)
```

Step-40 Top 5 Drop Location

```
print("="*60)
print("TOP 5 DROP LOCATION")
print("="*60)

print(
    drop.head()
)
```

Step-41 Daily Revenue

```
print("="*60)
print("DAILY REVENUE")
print("="*60)
```

```
daily = df.groupby(
    "Date"
) ["Fare (₹)"].sum()
```

```
print(daily)
```

Step-42 Daily Booking

```
print("="*60)
print("DAILY BOOKING")
print("="*60)
daily_booking = df.groupby(
    "Date"
) ["Booking ID"].count()
print(daily_booking)
```

Step-43 Completed Ride

```
completed = df[  
df["Booking Status"]=="Completed"  
]
```

```
print("="*60)  
print("COMPLETED RIDE")  
print("="*60)
```

```
print(completed)
```

Step-44 Completed Revenue

```
print("="*60)
print("COMPLETED REVENUE")
print("="*60)

print(
    completed["Fare (₹)"].sum()
)
```

Step-45 Cancelled Ride

```
cancelled = df[
    df["Booking Status"]=="Cancelled"
]

print("="*60)
print("CANCELLED RIDE")
print("="*60)

print(cancelled)
```

Step-46 Cancelled Count

```
print("="*60)
print("TOTAL CANCELLED")
print("="*60)

print(
cancelled.shape[0]
)
```

Step-47 Ongoing Ride

```
ongoing = df[
df["Booking Status"]=="Ongoing"
]

print("="*60)
print("ONGOING RIDE")
print("="*60)

print(ongoing)
```

Step-48 Ongoing Count

```
print("="*60)
print("TOTAL ONGOING")
print("="*60)
print(
ongoing.shape[0]
)
```

Step-49 Top 10 Expensive Rides

```
print("="*60)
print("TOP 10 EXPENSIVE RIDES")
print("="*60)
```

```
top10 = df.sort_values(
by="Fare (₹)",
ascending=False
)
```

```
print(
top10.head(10)
)
```

Step-50 Lowest 10 Fare

```
print("="*60)
print("LOWEST 10 FARE")
print("="*60)

low10 = df.sort_values(
by="Fare (₹) "
)

print(
low10.head(10)
)
```

Part-2 Complete

Part-3 Matplotlib Visualization cover :

- Pie Chart
- Bar Chart
- Horizontal Bar Chart
- Line Chart
- Histogram
- Revenue Chart
- Captain Performance Chart
- Payment Mode Chart
- Pickup Location Chart
- Daily Revenue Trend

- Aur saare charts ko charts/ folder me PNG format me save karna.

- **Part-3 : Data Visualization (Matplotlib)**

- Step-51 Import Library**

- ```
import matplotlib.pyplot as plt
import os
```

- ```
# Create charts folder if it doesn't exist
if not os.path.exists("charts"):
    os.makedirs("charts")
```

Step-52 Pie Chart (Booking Status)

```
status = df["Booking Status"].value_counts()

plt.figure(figsize=(6, 6))

plt.pie(
    status,
    labels=status.index,
    autopct="%1.1f%%",
    startangle=90
)

plt.title("Booking Status")

plt.savefig("charts/booking_status.png")

plt.show()
```

Step-53 Vehicle Type Bar Chart

```
vehicle = df["Vehicle Type"].value_counts()

plt.figure(figsize=(6,5))

plt.bar(vehicle.index, vehicle.values)

plt.title("Vehicle Wise Booking")

plt.xlabel("Vehicle")

plt.ylabel("Bookings")

plt.savefig("charts/vehicle_type.png")

plt.show()
```

Step-54 Payment Mode Chart

```
payment = df["Payment Mode"].value_counts()

plt.figure(figsize=(6,5))

plt.bar(payment.index, payment.values)

plt.title("Payment Mode")

plt.xlabel("Payment")

plt.ylabel("Booking")

plt.savefig("charts/payment_mode.png")

plt.show()
```

Step-55 Revenue By Vehicle

```
. vehicle_revenue = df.groupby(  
    "Vehicle Type"  
)[ "Fare (₹)" ].sum()  
  
plt.figure(figsize=(6,5))  
  
plt.bar(  
    vehicle_revenue.index,  
    vehicle_revenue.values  
)  
plt.title("Revenue By Vehicle")  
plt.xlabel("Vehicle")  
plt.ylabel("Revenue")  
  
plt.savefig("charts/revenue_vehicle.png")  
  
plt.show()
```

Step-56 Revenue By Captain

```
captain = df.groupby(
    "Captain Name"
)["Fare (₹)"].sum()
plt.figure(figsize=(8, 5))
plt.bar(
    captain.index,
    captain.values
)

plt.title("Captain Performance")

plt.xlabel("Captain")
plt.ylabel("Revenue")
plt.savefig("charts/captain.png")

plt.show()
```

Step-57 Daily Revenue Line Chart

```
daily = df.groupby(  
    "Date"  
)["Fare (₹)"].sum()  
plt.figure(figsize=(12, 5))  
plt.plot(  
    daily.index,  
    daily.values,  
    marker="o"  
)  
plt.xticks(rotation=90)  
plt.title("Daily Revenue")  
plt.xlabel("Date")  
plt.ylabel("Revenue")  
plt.savefig("charts/revenue.png")  
  
plt.show()
```

Step-58 Fare Histogram

```
plt.figure(figsize=(8,5))
plt.hist(
    df["Fare (₹)"],
    bins=10
)
plt.title("Fare Distribution")
plt.xlabel("Fare")
plt.ylabel("Frequency")
plt.savefig("charts/fare_histogram.png")
plt.show()
```

Step-59 Pickup Location Chart

```
pickup = df["Pickup Location"].value_counts()

plt.figure(figsize=(10, 5))

plt.bar(
    pickup.index,
    pickup.values
)

plt.xticks(rotation=45)

plt.title("Pickup Location")

plt.savefig("charts/pickup_location.png")

plt.show()
```

Step-60 Drop Location Chart

```
drop = df["Drop Location"].value_counts()

plt.figure(figsize=(10, 5))

plt.bar(
    drop.index,
    drop.values
)

plt.xticks(rotation=45)

plt.title("Drop Location")

plt.savefig("charts/drop_location.png")

plt.show()
```

Step-61 Booking Status Horizontal Chart

```
status = df["Booking Status"].value_counts()

plt.figure(figsize=(7, 5))

plt.barh(
    status.index,
    status.values
)

plt.title("Booking Status")

plt.savefig("charts/status_horizontal.png")

plt.show()
```

Step-62 Captain Trips Chart

```
trip = df["Captain Name"].value_counts()

plt.figure(figsize=(8, 5))

plt.bar(
    trip.index,
    trip.values
)
plt.title("Captain Trips")
plt.xlabel("Captain")

plt.ylabel("Trips")

plt.savefig("charts/captain_trip.png")

plt.show()
```

Step-63 Payment Pie Chart

```
payment = df["Payment Mode"].value_counts()

plt.figure(figsize=(6, 6))

plt.pie(
    payment,
    labels=payment.index,
    autopct="%1.1f%%"
)

plt.title("Payment Mode")

plt.savefig("charts/payment_pie.png")

plt.show()
```

Step-64 Vehicle Pie Chart

```
vehicle = df["Vehicle Type"].value_counts()

plt.figure(figsize=(6, 6))

plt.pie(
    vehicle,
    labels=vehicle.index,
    autopct="%1.1f%%"
)

plt.title("Vehicle Type")

plt.savefig("charts/vehicle_pie.png")

plt.show()
```

Step-65 Top 10 Revenue Chart

```
top = df.sort_values(  
    by="Fare (₹)",  
    ascending=False  
) .head(10)  
  
plt.figure(figsize=(10,5))  
  
plt.bar(  
    top["Booking ID"],  
    top["Fare (₹)"]  
)  
plt.xticks(rotation=45)  
plt.title("Top 10 Revenue Booking")  
  
plt.savefig("charts/top10_booking.png")  
  
plt.show()
```

Step-66 Daily Booking Chart

```
booking = df.groupby(  
    "Date"  
)["Booking ID"].count()  
  
plt.figure(figsize=(12, 5))  
  
plt.plot(  
    booking.index,  
    booking.values,  
    marker="o"  
)  
  
plt.xticks(rotation=90)  
  
plt.title("Daily Booking")  
plt.savefig("charts/daily_booking.png")  
  
plt.show()
```

Step-67 Scatter Plot

```
plt.figure(figsize=(8,5))

plt.scatter(
    range(len(df)),
    df["Fare (₹)"]
)

plt.title("Fare Scatter Plot")

plt.xlabel("Booking")

plt.ylabel("Fare")

plt.savefig("charts/scatter.png")

plt.show()
```

Step-68 Box Plot

```
plt.figure(figsize=(6,5))

plt.boxplot(
    df["Fare (₹)"]
)

plt.title("Fare Box Plot")

plt.savefig("charts/boxplot.png")

plt.show()
```

Step-69 Total Charts Generated

```
print("="*60)
print("All Charts Created Successfully")
print("="*60)
```

```
print("""
booking_status.png
vehicle_type.png
payment_mode.png
revenue_vehicle.png
captain.png
revenue.png
fare_histogram.png
pickup_location.png
drop_location.png
status_horizontal.png
captain_trip.png
payment_pie.png
vehicle_pie.png
```

```
top10_booking.png  
daily_booking.png  
scatter.png  
boxplot.png  
""")
```

Output (charts/ Folder)

charts/

- booking_status.png
- vehicle_type.png
- payment_mode.png
- revenue_vehicle.png
- captain.png
- revenue.png
- fare_histogram.png
- pickup_location.png
- drop_location.png
- status_horizontal.png
- captain_trip.png
- payment_pie.png
- vehicle_pie.png
- top10_booking.png
- daily_booking.png

└─ scatter.png
└─ boxplot.png

Part-4 (Final Project) :

- Report Generation (report.txt)
- Clean Dataset Export (clean_booking.csv)
- Monthly Summary
- Captain Performance Report
- Vehicle-wise Summary
- Payment Summary
- Top Customers
- Final Console Dashboard
- Project Conclusion

Part-4 (Final Project)

Assumption: df Part-1 me load ho chuka hai.

Step-70 Save Clean Dataset

```
# Save cleaned dataset  
  
df.to_csv("clean_booking.csv", index=False)  
  
print("Clean Dataset Saved Successfully")
```

Step-71 Total Booking

```
total_booking = len(df)  
  
print("Total Booking :", total_booking)
```

Step-72 Total Revenue

```
total_revenue = df["Fare (₹)"].sum()  
  
print("Total Revenue :", total_revenue)
```

Step-73 Average Fare

```
average_fare = df["Fare (₹)"].mean()  
  
print("Average Fare :", average_fare)
```

Step-74 Highest Fare

```
highest_fare = df["Fare (₹)"].max()  
  
print("Highest Fare :", highest_fare)
```

Step-75 Lowest Fare

```
lowest_fare = df["Fare (₹)"].min()  
  
print("Lowest Fare :", lowest_fare)
```

Step-76 Total Completed Booking

```
completed = df[df["Booking Status"]=="Completed"]  
  
print("Completed Booking :", len(completed))
```

Step-77 Total Cancelled Booking

```
cancelled = df[df["Booking Status"]=="Cancelled"]  
  
print("Cancelled Booking :", len(cancelled))
```

Step-78 Total Ongoing Booking

```
ongoing = df[df["Booking Status"]=="Ongoing"]  
  
print("Ongoing Booking :", len(ongoing))
```

Step-79 Vehicle Wise Summary

```
vehicle_summary = df.groupby("Vehicle Type").agg({  
    "Booking ID": "count",  
    "Fare (₹) ": "sum"  
})  
  
print(vehicle_summary)
```

Step-80 Captain Wise Summary

```
captain_summary = df.groupby("Captain Name").agg({  
    "Booking ID": "count",  
    "Fare (₹)": "sum"  
})  
  
print(captain_summary)
```

Step-81 Payment Summary

```
payment_summary = df.groupby("Payment Mode").agg({  
    "Booking ID": "count",  
    "Fare (₹)": "sum"  
})  
  
print(payment_summary)
```

Step-82 Pickup Summary

```
pickup_summary = df["Pickup Location"].value_counts()  
  
print(pickup_summary)
```

Step-83 Drop Summary

```
drop_summary = df["Drop Location"].value_counts()  
  
print(drop_summary)
```

Step-84 Top 10 Revenue Booking

```
top10 = df.sort_values(  
    by="Fare (₹)",  
    ascending=False  
) .head(10)  
  
print(top10)
```

Step-85 Lowest 10 Fare Booking

```
lowest10 = df.sort_values(  
    by="Fare (₹) "  
).head(10)  
print(lowest10)
```

Step-86 Top Captain

```
top_captain = df.groupby(  
    "Captain Name"  
).["Fare (₹)"].sum().sort_values(  
    ascending=False  
)  
print(top_captain)
```

Step-87 Top Pickup Location

```
top_pickup = df["Pickup Location"].value_counts()  
print(top_pickup.head())
```

Step-88 Top Drop Location

```
top_drop = df["Drop Location"].value_counts()  
  
print(top_drop.head())
```

Step-89 Save Vehicle Summary

```
vehicle_summary.to_csv(  
    "vehicle_summary.csv"  
)
```

Step-90 Save Captain Summary

```
captain_summary.to_csv(  
    "captain_summary.csv"  
)
```

Step-91 Save Payment Summary

```
payment_summary.to_csv(  
    "payment_summary.csv"  
)
```

Step-92 Report Generation

```
with open("report.txt", "w") as f:

    f.write("="*60+"\n")

    f.write("RAPIDO BOOKING REPORT\n")

    f.write("="*60+"\n\n")

    f.write("Total Booking : "+str(total_booking))

    f.write("\n")

    f.write("Completed Booking : "+str(len(completed)))

    f.write("\n")

    f.write("Cancelled Booking : "+str(len(cancelled)))
```

```
f.write("\n")
```

```
f.write("Ongoing Booking : "+str(len(ongoing)))
```

```
f.write("\n")
```

```
f.write("Total Revenue : "+str(total_revenue))
```

```
f.write("\n")
```

```
f.write("Average Fare : "+str(round(average_fare,2)))
```

```
f.write("\n")
```

```
f.write("Highest Fare : "+str(highest_fare))
```

```
f.write("\n")
```

```
f.write("Lowest Fare : "+str(lowest_fare))
```

```
f.write("\n")
```

```
print("Report Created Successfully")
```

Step-93 Final Console Dashboard

```
print("\n")

print("="*70)

print("          RAPIDO BOOKING ANALYSIS DASHBOARD")

print("="*70)

print("Total Booking          :", total_booking)

print("Completed Booking       :", len(completed))

print("Cancelled Booking        :", len(cancelled))

print("Ongoing Booking          :", len(ongoing))

print("Total Revenue            :", total_revenue)
```

```
print("Average Fare          :", round(average_fare,2))  
print("Highest Fare         :", highest_fare)  
print("Lowest Fare          :", lowest_fare)  
print("="*70)
```

Step-94 Export Everything

```
print("Files Created Successfully")
```

```
print("""
```

```
clean_booking.csv
```

```
vehicle_summary.csv
```

```
captain_summary.csv
```

```
payment_summary.csv
```

```
report.txt
```

```
charts/
```

```
""")
```

Final Project Folder

- Rapido_Project/

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|— rapido_booking.csv

|— clean_booking.csv

|— analysis.py

|— vehicle_summary.csv

|— captain_summary.csv

|— payment_summary.csv

|— report.txt

|

|— charts/

| booking_status.png

| payment_mode.png

| revenue.png

| vehicle_type.png

| revenue_vehicle.png

| captain.png

| fare_histogram.png

| pickup_location.png

- | drop_location.png
- | status_horizontal.png
- | payment_pie.png
- | vehicle_pie.png
- | top10_booking.png
- | daily_booking.png
- | scatter.png
- | boxplot.png

Project Complete

